Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/06/2024
PI Name:	Turkoglu, Danyal Ph.D.		
Project Title:	Radioisotope to Enable X-Ray Based Inflight Space Radiology		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98199	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2023 HERO NNJ23ZSA001N-OMNIBUS : NASA Human Research Program Omnibus Opportunity
Start Date:	03/01/2024	**End Date:**	09/30/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Lemery, Jay	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	USNC entered Chapter 11 proceedings and consequently the PI, Danyal Turkoglu, has left the company. John Krueger is filling in to complete this Task Book update.		
COI Name (Institution):	Choi, John M.D., Ph.D. (University of Southern California) Fraser, Scott Ph.D. (University of Southern California)		
Grant/Contract No.:	80NSSC24K0949		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Long-term manned space missions, such as to Mars, are expected to require inflight medical capabilities. Inflight space radiology is critical to proper diagnosis. As an example, a left internal jugular venous thrombosis was diagnosed by ultrasound and treated with anticoagulation, all while aboard the International Space Station (ISS). However, a thrombus in the pulmonary or cerebral arteries is best diagnosed with X-ray (XR) based computed tomography (CT). Thus, we propose a method to enable infight XR based imaging, overcoming the power, cost, and mass limitations. Using radioisotopes with inherent XR emission through radioactive decay, our proposed XR source will require no external electrical power to generate photons, compared to the kilowatts needed for a terrestrial XR or CT machine. This inflight medical capability can reduce the risk of adverse health outcomes due to medical conditions that occur in mission, such as, but not limited to, thrombi, renal stones, or fractures. Radioisotope-based radiography was originally motivated by portability, but has become obsolete due to battery technology, which would still be impractical for space. We propose to adapt this technique with modern intensifying screens and digital image technology (versus analog film). We will primarily investigate the requirements for a suitable radioisotope, including manufacturability, using our existing expertise in modelling and simulation. We have manufactured and irradiated a ceramic target (thulia precursor + alumina encapsulation) and activated it to greater than 100 millicuries Tm-170, that may provide an intensity suitable for XR imaging. Further work is needed to adapt a design optimized for heat production to a design optimized for escaping XR emissions at clinically relevant energies. We aim for a custom form factor that provides a practical path from ground-based tests to flight qualification.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Radioisotope-based X-ray (XR) imaging enables new inflight capabilities for diagnosis, and potentially for treatment, of astronaut medical conditions that might prevent future long-term manned mission success.
Task Progress:	For this project, we evaluated radioisotopes with intrinsic radiation suitable for medical imaging, including Co-57, Tm-170, and Cs-137. Tm-170 was the original focus of the project because it emits radiation at clinically relevant energies (e.g., 84 keV photopeak), and USNC has a license to use it. However, Tm-170 was not available in large activities. Co-57 was a convenient stand-in for Tm-170, and it was convenient to use because it can be possessed in activities of 100 uCi or less without a radioactive materials license. These sources were combined with several detection techniques, including a dark box containing a scintillation screen and camera, a lens tube attached to a DSLR camera with the source attached to Gadox phosphor, and direct coupling of the source to a dental x-ray sensor. Images were successfully captured using Tm-170 and the dark box configuration, as well as Co-57 and Cs-137 and the lens tube. Cs-137 was also able to produce an image using the dental sensor, but the image was vastly improved by using a clinical x-ray source.
Bibliography Type:	Description: (Last Updated: 07/29/2025)